USN | | | | | | | | | | [22CSE306]

SECOND SEMESTER B.Tech EXAMINATION, APRIL - 2023

OBJECT ORIENTED PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

PART-A **05 X 02=10**

1.	**a.**	List the principles of software engineering.	L-1	CO1	**02**
	b.	Highlight the difference between data hiding and encapsulation.	L-1	CO2	**02**
	c.	List the different types of inheritance.	L-1	CO3	**02**
	d.	Define polymorphism with an example.	L-1	CO5	**02**
	e.	Define exceptions in Java.	L-1	CO6	**02**
		PART-B		**06X 05=30**	
2.	**a.**	Discuss the difference between class and object in Java.	L-2	CO1	**05**
		OR			
	b.	Define static keyword with an example.	L-2	CO1	**05**
3.	**a.**	List the different access modifiers in Java with an example for each.	L-2	CO2	**05**
		OR			
	b.	Define POJO class with an example.	L-2	CO2	**05**
4.	**a.**	Define polymorphism and its types with an example.	L-2	CO5	**05**
		OR			
	b.	How is Inheritance useful to achieve Polymorphism in Java? List the difference between inheritance and polymorphism in Java.	L-2	CO5	**05**
5.	**a.**	Discuss the difference between exception and error.	L-2	CO6	**05**
		OR			
	b.	Discuss the difference between throw and throws in Java.	L-2	CO6	**05**
6.	**a.**	Write a Java program to illustrate checked and unchecked exceptions.	L-3	CO6	**05**
		OR			
	b.	Define user-defined exception in Java with an example.	L-2	CO5	**05**

7.	**a.**	Write a method for computing x^y doing repetitive multiplication. X and y are of type integer and are to be given as command line arguments.	L-3	CO1	**05**
		OR			
	b.	Describe abstract class called Shape which has three subclasses say Triangle, Rectangle, Circle. Define one method area() in the abstract class and override this area() in these three subclasses to calculate for specific object. Area of Triangle = (1/2) x base x height Area of Circle = pi x radius x radius Area of Rectangle = Length x Breadth	L-3	CO4	**05**
		PART-C		**06 X 10=60**	
8.	**a.**	Write a Java program to demonstrate the use of static variable and static method.	L-3	CO1	**10**
		OR			
	b.	Write a class that contains the following two methods: to Convert from celsius to fahrenheit and to convert from fahrenheit to Celsius. Celsius = [(Fahrenheit – 32) x 5]/9	L-3	CO1	**10**
9.	**a.**	Explain Exception handling in JAVA. Write an application that generates custom exception if any value from its command line arguments is negative.	L-3	CO6	**10**
		OR			
	b.	Recall what is package? Explain how to create user defined package in java with example.	L-3	CO2	**10**
10.	**a.**	Write a java program to implement multilevel inheritance with 4 levels of hierarchy.	L-3	CO3	**10**
		OR			
	b.	What is the importance of **super** and **this** keyword in inheritance? Illustrate with suitable example.	L-3	CO3	**10**
11.	**a.**	Consider a parent class as living thing. Living things exist on the planet in the forms of human beings, animals, plants, bacteria,	L-3	CO5	**10**

		etc. These are the child classes inherited from the parent class, i.e., living things. Construct a Java program using the concept of polymorphism. [Clue: The parent class *Livingthings* has the method *live*(). Subclasses of Livingthings i.e., *Animals*, *Humanbeing*, and *Plants* have their way of living. By the principle of inheritance and polymorphism, each subclass can define its way of living with the help of the *live*() method.]			
		OR			
	b.	Explain in detail method overriding with suitable example.	L-3	CO4	**10**
12.	**a.**	What is exception? Write a program to accept a password from the user and throw "Authentication Failure" exception if the password is incorrect.	L-3	CO6	**10**
		OR			
	b.	Show the use of finally statement with an program for any two of the following cases. **i.** Case 1: When an exception does not occur **ii.** Case 2: When an exception occurs but not handled by the catch block **iii.** Case 3: When an exception occurs and is handled by the catch block	L-3	CO6	**10**
13.	**a.**	Declare a class called author having author_name as private data member. Extend author class to have two sub classes called book_publication & paper_publication. Each of these classes have private member called title. Show usage of dynamic method dispatch (dynamic polymorphism) to display book or paper publications of a given author. Use command line arguments for inputting data.	L-3	CO5	**10**
		OR			
	b.	The abstract vegetable class has three subclasses named Potato, Brinjal and Tomato. Write a java program that demonstrates how to establish this class hierarchy. Declare one instance variable of type String that indicates the color of a vegetable. Create and display instances of these objects. Override the toString() method of object to return a string with the name of vegetable and its color.	L-3	CO4	**10**